

# Assignment 1

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## 1 Bayesian Change-Point Model

Let  $D = \{x_1, x_2, \dots, x_N\}$  be the observed sequence of count data. The observations are assumed to follow a Poisson model with a single unknown change point  $m$ . Before the change point, the observations are generated with rate  $\lambda_1$ , while after the change point they are generated with rate  $\lambda_2$ .

The change point is restricted to the set

$$\mathcal{M} = \{n_{\min}, n_{\min} + 1, \dots, N - n_{\min}\},$$

with  $n_{\min} = 5$  and  $N$  the length of the dataset, in order to avoid degenerate segments containing too few observations.

The prior distributions are  $\lambda_1 \sim \text{Gamma}(\alpha, \beta)$  and  $\lambda_2 \sim \text{Gamma}(\alpha, \beta)$ , with density parameterized by shape and rate:

$$p(\lambda \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}, \quad \lambda > 0,$$

We use also  $m \sim \text{Uniform}(\mathcal{M})$  and we assign the given values to the prior distributions:  $\alpha = 1$ ,  $\beta = 0.1$ .

### 1.1 Likelihood Function

Conditionally on  $(\lambda_1, \lambda_2, m)$ , the observations are independent. Therefore, the likelihood can be written as

$$p(D \mid \lambda_1, \lambda_2, m) = \prod_{i=1}^m \frac{e^{-\lambda_1} \lambda_1^{x_i}}{x_i!} \prod_{i=m+1}^N \frac{e^{-\lambda_2} \lambda_2^{x_i}}{x_i!}.$$

Grouping together all terms depending on the parameters gives

$$p(D \mid \lambda_1, \lambda_2, m) = \left( \prod_{i=1}^m \frac{1}{x_i!} \right) e^{-m\lambda_1} \lambda_1^{\sum_{i=1}^m x_i} e^{-(N-m)\lambda_2} \lambda_2^{\sum_{i=m+1}^N x_i}.$$

Defining the cumulative sums

$$S_m = \sum_{i=1}^m x_i, \quad S_N = \sum_{i=1}^N x_i,$$

the likelihood becomes

$$p(D \mid \lambda_1, \lambda_2, m) \propto \lambda_1^{S_m} e^{-m\lambda_1} \lambda_2^{S_N - S_m} e^{-(N-m)\lambda_2},$$

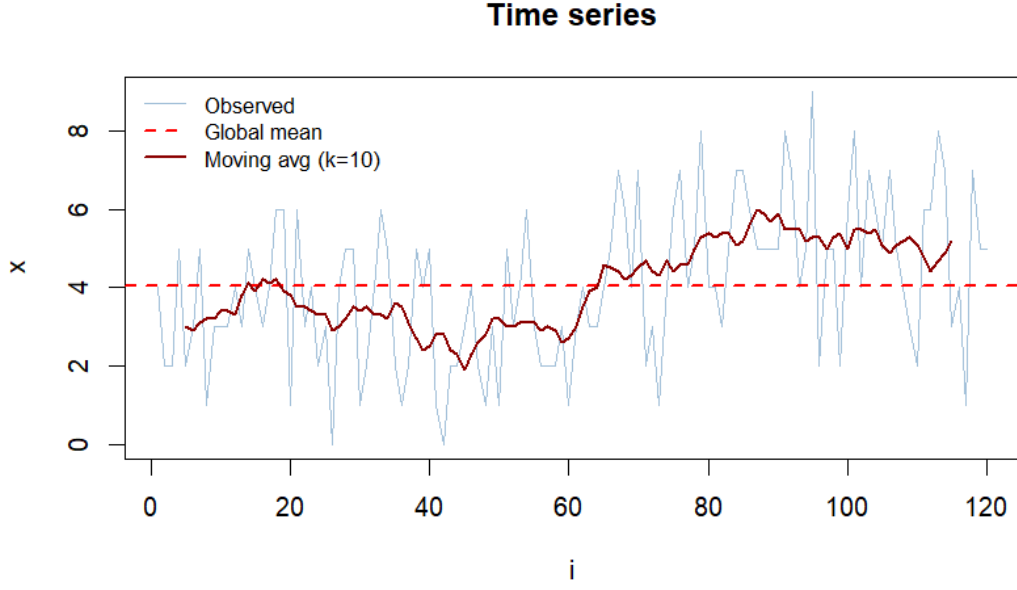


Figure 1: Time series of the dataset

## 1.2 Posterior Distribution

Using Bayes theorem, we have:

$$p(\lambda_1, \lambda_2, m \mid D) \propto p(D \mid \lambda_1, \lambda_2, m)p(\lambda_1)p(\lambda_2)p(m).$$

Substituting the likelihood and prior distributions yields

$$p(\lambda_1, \lambda_2, m \mid D) \propto \lambda_1^{S_m} e^{-m\lambda_1} \lambda_2^{S_N - S_m} e^{-(N-m)\lambda_2} \lambda_1^{\alpha-1} e^{-\beta\lambda_1} \lambda_2^{\alpha-1} e^{-\beta\lambda_2}.$$

Collecting terms gives

$$p(\lambda_1, \lambda_2, m \mid D) \propto \lambda_1^{S_m + \alpha - 1} e^{-(m+\beta)\lambda_1} \lambda_2^{S_N - S_m + \alpha - 1} e^{-(N-m+\beta)\lambda_2}.$$

Since the prior for  $m$  is uniform on  $\mathcal{M}$ , it contributes only with a constant factor and can be omitted from the proportional expression.

## 1.3 Conditional Posterior Distributions of $\lambda_1$ and $\lambda_2$

Conditioning on  $m$ , the posterior distribution factorizes into two Gamma distributions.

For  $\lambda_1$ ,

$$p(\lambda_1 \mid m, D) \propto \lambda_1^{S_m + \alpha - 1} e^{-(m+\beta)\lambda_1}.$$

This is recognized as the kernel of a Gamma distribution:

$$\lambda_1 \mid m, D \sim \text{Gamma}(\alpha + S_m, \beta + m).$$

Similarly,

$$\lambda_2 \mid m, D \sim \text{Gamma}(\alpha + S_N - S_m, \beta + N - m).$$

## 1.4 Conditional Posterior Distribution of $m$

Conditioning on  $(\lambda_1, \lambda_2)$ , the posterior distribution of  $m$  is proportional to the likelihood:

$$p(m \mid \lambda_1, \lambda_2, D) \propto p(D \mid \lambda_1, \lambda_2, m).$$

Using the likelihood expression,

$$p(m \mid \lambda_1, \lambda_2, D) \propto \lambda_1^{S_m} e^{-m\lambda_1} \lambda_2^{S_N - S_m} e^{-(N-m)\lambda_2}, \quad m \in \mathcal{M}.$$

Taking logarithms gives

$$\log p(m \mid \lambda_1, \lambda_2, D) = S_m \log \lambda_1 - m\lambda_1 + (S_N - S_m) \log \lambda_2 - (N - m)\lambda_2 + C,$$

where  $C$  is a constant independent of  $m$ .

The parameter  $m$  is discrete because it represents the index of the observation at which the rate changes. Consequently, its log-posterior distribution is defined over the finite discrete set  $\mathcal{M}$ , and probabilities must be normalized by summing over all admissible values:

$$p(m \mid \lambda_1, \lambda_2, D) = \frac{w(m)}{\sum_{k \in \mathcal{M}} w(k)},$$

where

$$w(m) = \lambda_1^{S_m} e^{-m\lambda_1} \lambda_2^{S_N - S_m} e^{-(N-m)\lambda_2}.$$

In practice, to avoid numerical underflow, the probabilities are computed on the logarithmic scale and normalized after subtracting the maximum log-probability.

## 1.5 Why MCMC is Appropriate

Even though the conditional posterior distributions of  $\lambda_1$  and  $\lambda_2$  are available analytically, the joint posterior distribution  $p(\lambda_1, \lambda_2, m \mid D)$  is still difficult to analyze directly.

The main difficulty comes from the dependence between the continuous parameters  $(\lambda_1, \lambda_2)$  and the discrete change point  $m$ . We have that the posterior distribution does not factorize into independent components and the parameter  $m$  introduces a combinatorial structure. Also, posterior expectations and probabilities generally require a sum over all possible values of  $m$  together with integration over  $\lambda_1$  and  $\lambda_2$ .

Markov chain Monte Carlo methods allow us to generate dependent samples directly from the posterior distribution by iteratively sampling from simpler conditional distributions.

In this model, Gibbs sampling is particularly convenient since we are using Gamma distributions that can be sampled exactly and the conditional distribution of  $m$  is discrete, so it can be sampled by evaluating normalized probabilities over the finite set  $\mathcal{M}$ .

## 2 Gibbs Sampling

Since the full joint posterior distribution  $p(\lambda_1, \lambda_2, m \mid D)$  cannot be sampled directly in closed form, we use a Gibbs sampler to generate dependent samples from the posterior distribution.

The Gibbs sampler uses the fact that all conditional posterior distributions are available explicitly.

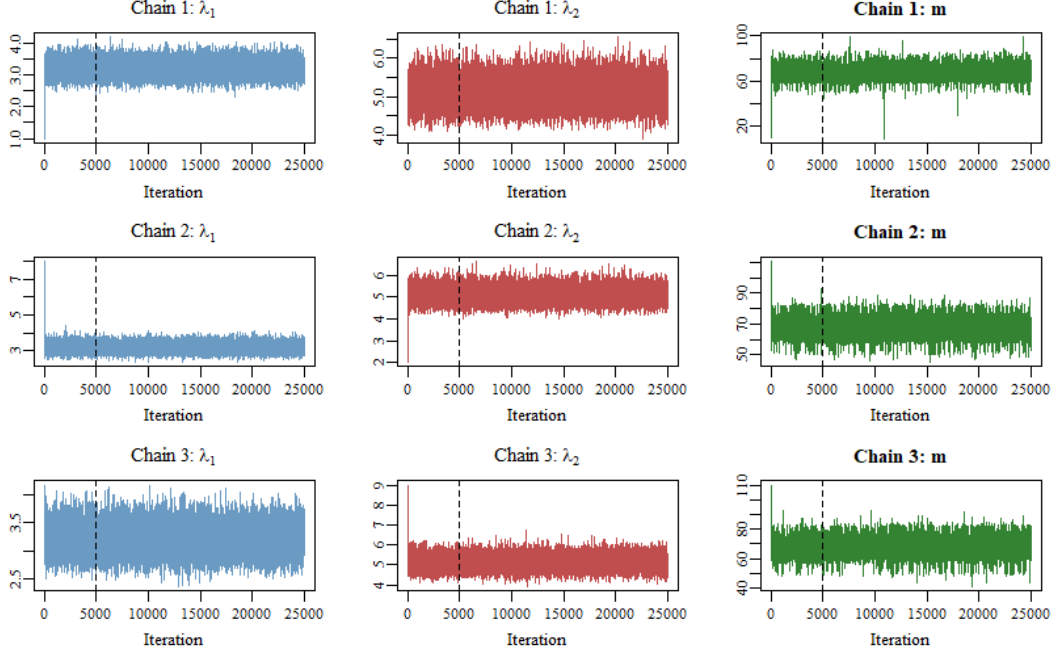


Figure 2: Trace of the three parameters of the three chains. The dotted line indicates burn-in

## 2.1 Gibbs Sampling Updates

Using the results derived in Section 1, one iteration of the Gibbs sampler consists of the following three steps. First we update  $\lambda_1$ :

$$\lambda_1^{(t)} \sim \text{Gamma} \left( \alpha + S_{m^{(t-1)}}, \beta + m^{(t-1)} \right),$$

Then similarly  $\lambda_2$ :

$$\lambda_2^{(t)} \sim \text{Gamma} \left( \alpha + S_N - S_{m^{(t-1)}}, \beta + N - m^{(t-1)} \right).$$

In the end we update  $m$ . Because the posterior distribution of  $m$  may involve very small probabilities, the computation is performed on the logarithmic scale:

$$\log w(m) = S_m \log \lambda_1 - m \lambda_1 + (S_N - S_m) \log \lambda_2 - (N - m) \lambda_2.$$

To improve numerical stability, the maximum log-probability is subtracted before exponentiation and then, the normalized probabilities are obtained as

$$p(m \mid \lambda_1, \lambda_2, D) = \frac{\tilde{w}(m)}{\sum_{k \in \mathcal{M}} \tilde{w}(k)}.$$

where  $\tilde{w}(m) = \exp(\log w(m) - \max_k \log w(k))$ . Finally, a new value of  $m$  is sampled from this discrete distribution.

## 2.2 Multiple Chains and Convergence Assessment

In order to verify the convergence of the algorithm, three independent Markov chains were initialized from different starting values  $(\lambda_1^{(0)}, \lambda_2^{(0)}, m^{(0)})$ .

The Gibbs sampler was run for a sufficiently large number of iterations ( $n_{iter} = 25000$ ) in order to obtain stable posterior estimates.

Traceplots for  $\lambda_1$ ,  $\lambda_2$ , and  $m$  are used to visually inspect convergence. The chain fluctuate around a stable region without exhibiting systematic trends or long transient phases.

The traceplots produced by the three chains showed that the chains converged toward similar posterior regions after the initial transient phase.

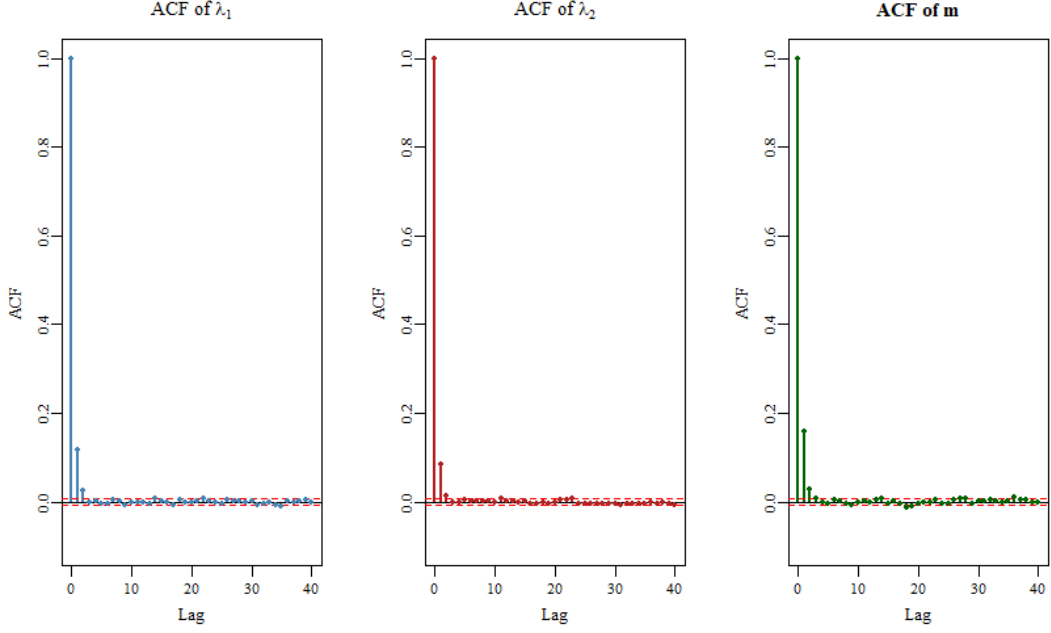


Figure 3: ACF of the three parameters

### 2.3 Burn-in Selection

The initial iterations of an MCMC algorithm may still be influenced by the starting values and therefore may not yet represent draws from the stationary posterior distribution. For this reason, a burn-in period was introduced. The burn-in length was selected by visually inspecting the traceplots and identifying the point after which the chains appeared stationary, the effect of initialization became negligible and the chains from different starting values overlapped consistently. Visually from fig. 2 we could choose for a shorter length but 5000 is a safer choice (and to have a nice summation of the three chains later on that can be inspected rapidly).

### 2.4 Autocorrelation and Effective Sample Size

Since Gibbs sampling produces dependent samples, consecutive observations in the chain are correlated. To quantify this dependence, autocorrelation functions (ACF) were computed for the three parameters.

Large autocorrelations imply that the chain explores the posterior space slowly, reducing the amount of effectively independent information contained in the simulated sample.

To evaluate sampling efficiency, the effective sample size (ESS) was estimated. The ESS measures the equivalent number of independent observations represented by the correlated MCMC sample.

If  $\rho_k$  denotes the recalibrated autocorrelation of the observable, the ESS is approximately given by

$$ESS = \frac{M}{1 + 2 \sum_{k=1}^{\infty} \rho_k},$$

where  $M$  is the total number of retained iterations after burn-in.

A larger ESS indicates more reliable posterior estimation and better mixing behavior.

$$ESS \lambda_1 = 46217.69 \quad ESS \lambda_2 = 50010.55 \quad ESS m = 43481.48$$

The ESS values obtained for the parameters are sufficiently large to support stable posterior inference.

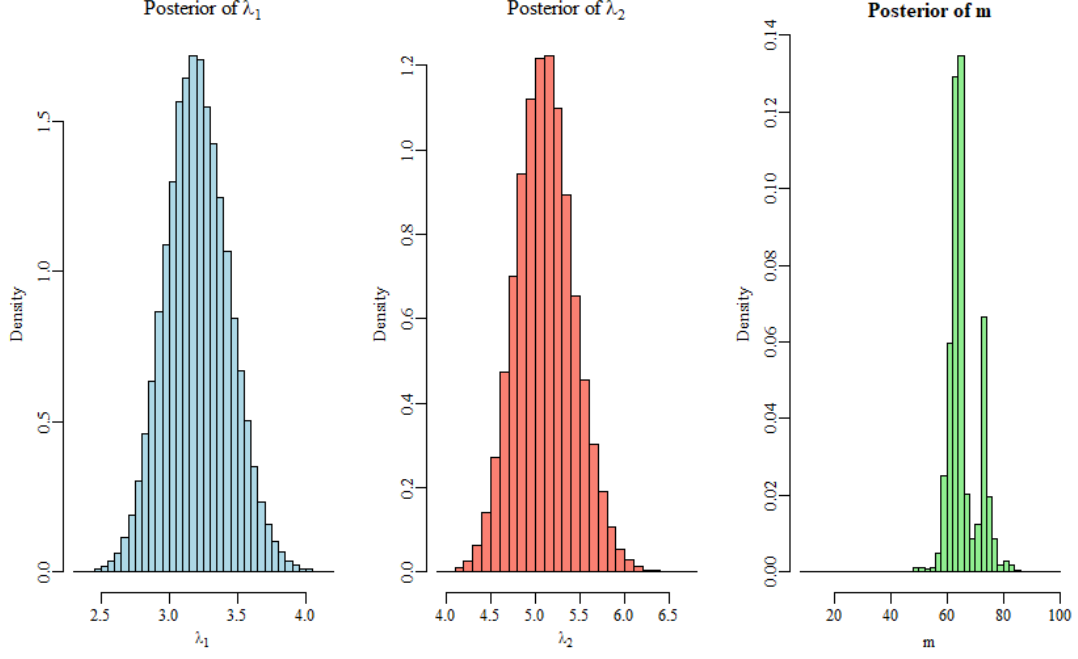


Figure 4: Posterior distributions of the three parameters

## 2.5 Posterior Summaries

Posterior summaries are computed using the retained Gibbs samples after burn-in removal.

| Parameter   | Mean  | Median | SD    | MAP   | CI 95%         |
|-------------|-------|--------|-------|-------|----------------|
| $\lambda_1$ | 3.207 | 3.200  | 0.234 | 3.196 | [2.762, 3.678] |
| $\lambda_2$ | 5.111 | 5.102  | 0.324 | 5.112 | [4.503, 5.774] |
| $m$         | 66.21 | 65.00  | 5.125 | 64.92 | [59.00, 77.00] |

Table 1: Posterior summaries for  $\lambda_1$ ,  $\lambda_2$ , and  $m$ .

The posterior mean is estimated as

$$\hat{\mathbb{E}}[\theta \mid D] = \frac{1}{M} \sum_{t=1}^M \theta^{(t)},$$

where  $\theta^{(t)}$  denotes the retained MCMC samples.

The posterior median and credible intervals are estimated using empirical quantiles of the sampled values.

The MAP estimate is approximated numerically from the empirical posterior density.

## 2.6 Posterior Probability $P(\lambda_2 > \lambda_1 \mid D)$

An important quantity of interest is the posterior probability that the post-change rate exceeds the pre-change rate:

$$P(\lambda_2 > \lambda_1 \mid D).$$

This probability was estimated directly from the Gibbs output as

$$\hat{P}(\lambda_2 > \lambda_1 \mid D) = \frac{1}{M} \sum_{t=1}^M \mathbf{1}(\lambda_2^{(t)} > \lambda_1^{(t)}),$$

where  $\mathbf{1}(\cdot)$  denotes the indicator function.

A value close to 1 would indicate strong posterior evidence that the event rate increased after the change point, while a value close to 0 would indicate evidence for a decrease. In our case, we have a value closer to 1, supported by the CI in the previous section.

### 3 Posterior Predictive Distribution

After estimating the posterior distribution of the parameters  $(\lambda_1, \lambda_2, m)$ , we now study the predictive distribution of a future observation. Let  $x_{\text{new}}$  denote the number of events observed in the next time interval after the end of the dataset.

The posterior predictive distribution quantifies uncertainty about future observations by integrating over the posterior uncertainty of the unknown parameters.

#### 3.1 Posterior Predictive Distribution

The posterior predictive distribution is defined as

$$p(x_{\text{new}} | D) = \sum_{m \in \mathcal{M}} \int \int p(x_{\text{new}} | \lambda_1, \lambda_2, m) p(\lambda_1, \lambda_2, m | D) d\lambda_1 d\lambda_2.$$

This expression averages the predictive distribution over the entire posterior distribution of the unknown parameters.

Since the future observation occurs after the final observed time point  $N$ , it belongs to the post-change regime. Consequently, conditional on the parameters, the future count follows  $x_{\text{new}} | \lambda_2 \sim \text{Poisson}(\lambda_2)$ .

So we can write

$$p(x_{\text{new}} | \lambda_1, \lambda_2, m) = p(x_{\text{new}} | \lambda_2) = \frac{e^{-\lambda_2} \lambda_2^{x_{\text{new}}}}{x_{\text{new}}!}.$$

and it follows that the posterior predictive distribution can be written as

$$p(x_{\text{new}} | D) = \sum_{m \in \mathcal{M}} \int \int \frac{e^{-\lambda_2} \lambda_2^{x_{\text{new}}}}{x_{\text{new}}!} p(\lambda_1, \lambda_2, m | D) d\lambda_1 d\lambda_2.$$

#### 3.2 Posterior Predictive Mean

The posterior predictive mean is defined as  $\mathbb{E}[x_{\text{new}} | D]$ . Using the law of total expectation,

$$\mathbb{E}[x_{\text{new}} | D] = \mathbb{E}[\mathbb{E}[x_{\text{new}} | \lambda_2, D] | D].$$

Since

$$x_{\text{new}} | \lambda_2 \sim \text{Poisson}(\lambda_2),$$

we have  $\mathbb{E}[x_{\text{new}} | \lambda_2] = \lambda_2$ .

Therefore,

$$\mathbb{E}[x_{\text{new}} | D] = \mathbb{E}[\lambda_2 | D].$$

Using the Gibbs samples,  $\{\lambda_2^{(1)}, \dots, \lambda_2^{(M)}\}$ , the posterior predictive mean can be estimated as

$$\widehat{\mathbb{E}}[x_{\text{new}} | D] = \frac{1}{M} \sum_{t=1}^M \lambda_2^{(t)} = 4.05.$$

This Monte Carlo estimator converges to the true posterior predictive mean as the number of retained samples increases.

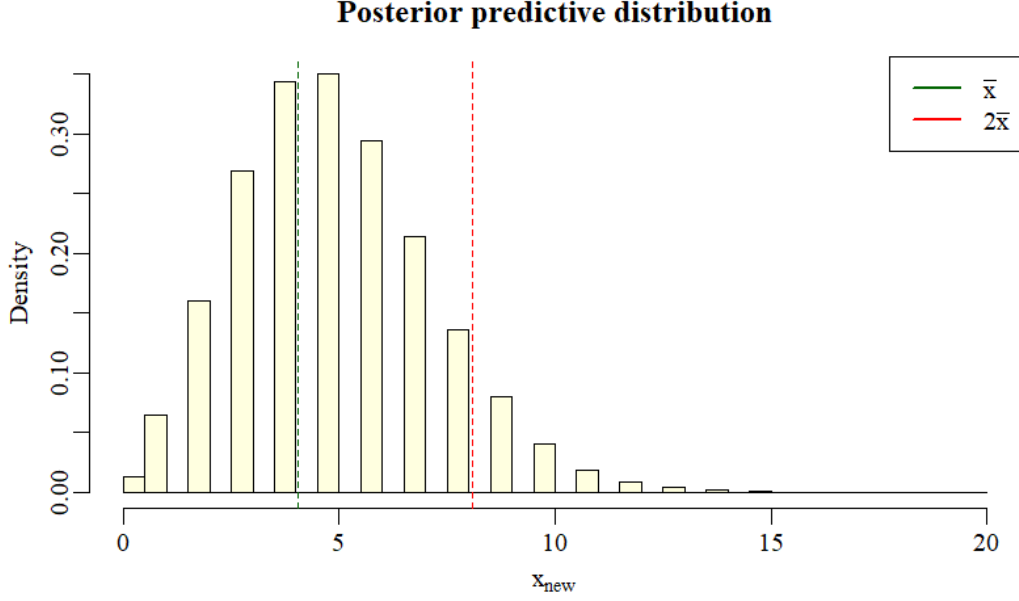


Figure 5: Posterior predictive with the empirical mean and the threshold equivalent to the double of the first value

### 3.3 Estimation of $P(x_{\text{new}} > 2\bar{x} \mid D)$

Let  $\bar{x}$  be the empirical mean of the observed data. We are interested in the posterior predictive probability  $P(x_{\text{new}} > 2\bar{x} \mid D)$ , which measures the probability of observing a future count larger enough than the historical average.

Using posterior predictive simulations, this probability can be estimated as

$$\hat{P}(x_{\text{new}} > 2\bar{x} \mid D) = \frac{1}{M} \sum_{t=1}^M \mathbf{1}(x_{\text{new}}^{(t)} > 2\bar{x}),$$

A large value would indicate that high counts are plausible under the posterior predictive distribution, while a small value would suggest that such events are unlikely.

What we obtain is  $P(x_{\text{new}} > 2\bar{x} \mid D) = 0,077$ .

## 4 Model Comparison

We now compare the one-change-point model  $M_1$  with the simpler no-change model  $M_0$ .

The purpose of this analysis is to determine whether the observed data provide evidence in favor of a structural change in the event rate.

The two competing models are:

$$M_0 : \quad x_i \mid \lambda \sim \text{Poisson}(\lambda), \quad i = 1, \dots, N,$$

and

$$M_1 : \quad \begin{cases} x_i \mid \lambda_1, m \sim \text{Poisson}(\lambda_1), & i = 1, \dots, m, \\ x_i \mid \lambda_2, m \sim \text{Poisson}(\lambda_2), & i = m + 1, \dots, N. \end{cases}$$

Both models use Gamma priors with the same hyperparameters:

$$\lambda \sim \text{Gamma}(\alpha, \beta), \quad \lambda_1, \lambda_2 \sim \text{Gamma}(\alpha, \beta).$$

The comparison is performed using the Bayes factor:

$$BF_{10} = \frac{p(D | M_1)}{p(D | M_0)},$$

#### 4.1 Marginal Likelihood under $M_0$

Under model  $M_0$ , all observations share the same Poisson rate  $\lambda$ .

The likelihood is

$$p(D | \lambda, M_0) = \prod_{i=1}^N \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}.$$

that gives

$$p(D | \lambda, M_0) = \left( \prod_{i=1}^N \frac{1}{x_i!} \right) e^{-N\lambda} \lambda^{S_N},$$

The prior distribution is

$$p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}.$$

The marginal likelihood is obtained by integrating out  $\lambda$ :

$$p(D | M_0) = \int p(D | \lambda, M_0) p(\lambda) d\lambda = \left( \prod_{i=1}^N \frac{1}{x_i!} \right) \frac{\beta^\alpha}{\Gamma(\alpha)} \int \lambda^{S_N+\alpha-1} e^{-(N+\beta)\lambda} d\lambda.$$

Using the Gamma integral identity,

$$\int_0^\infty x^{a-1} e^{-bx} dx = \frac{\Gamma(a)}{b^a},$$

we obtain

$$p(D | M_0) = \left( \prod_{i=1}^N \frac{1}{x_i!} \right) \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{\Gamma(S_N + \alpha)}{(N + \beta)^{S_N + \alpha}} \propto \frac{\Gamma(S_N + \alpha)}{(N + \beta)^{S_N + \alpha}},$$

up to factors common to both models.

#### 4.2 Marginal Likelihood under $M_1$

Under the change-point model, the marginal likelihood is

$$p(D | M_1) = \sum_{m \in \mathcal{M}} p(m) \int \int p(D | \lambda_1, \lambda_2, m) p(\lambda_1) p(\lambda_2) d\lambda_1 d\lambda_2.$$

Using the likelihood derived previously,

$$p(D | \lambda_1, \lambda_2, m) \propto \lambda_1^{S_m} e^{-m\lambda_1} \lambda_2^{S_N - S_m} e^{-(N-m)\lambda_2},$$

and combining it with the Gamma priors gives

$$\begin{aligned} p(D | M_1) &\propto \sum_{m \in \mathcal{M}} \int \lambda_1^{S_m + \alpha - 1} e^{-(m+\beta)\lambda_1} d\lambda_1 \\ &\quad \times \int \lambda_2^{S_N - S_m + \alpha - 1} e^{-(N-m+\beta)\lambda_2} d\lambda_2. \end{aligned}$$

Applying the Gamma integral separately to the two terms:

$$p(D | M_1) = \sum_{m \in \mathcal{M}} \left( \prod_{i=1}^N \frac{1}{x_i!} \right) \left( \frac{\beta^\alpha}{\Gamma(\alpha)} \right)^2 \cdot \frac{1}{N - 10} \frac{\Gamma(S_m + \alpha)}{(m + \beta)^{S_m + \alpha}} \cdot \frac{\Gamma(S_N - S_m + \alpha)}{(N - m + \beta)^{S_N - S_m + \alpha}}.$$

### 4.3 Bayes Factor

The Bayes factor comparing the two models is

$$BF_{10} = \frac{p(D \mid M_1)}{p(D \mid M_0)}.$$

Computations are done on the logarithmic scale:

$$\log BF_{10} = \log p(D \mid M_1) - \log p(D \mid M_0).$$

From the data we extrapolate a  $BF \approx 4 \times 10^3$ , giving a decisive evidence in favour of the  $M_1$  model.

## 5 Discussion

The Bayesian change-point analysis provides strong evidence that the event rate is not constant throughout the observed time series. Both the posterior inference obtained from the Gibbs sampler and the model comparison performed through the Bayes factor consistently support the existence of a structural change in the Poisson intensity.

### 5.1 Evidence for a Change Point

Several elements indicate the presence of a change point in the data.

First, the posterior distributions of the two Poisson rates are clearly separated. The posterior probability was estimated to be extremely close to 1, providing direct Bayesian evidence that the event rate increased after the change point. On top of that, the Bayes factor comparing the change-point model with the homogeneous model was estimated as  $BF_{10} \approx 4 \times 10^3$ .

The observed data strongly reject the hypothesis of a single constant Poisson rate over time.

### 5.2 Most Plausible Change-Point Location

The posterior distribution of the change point  $m$  was concentrated around  $\mathbb{E}[m \mid D] \approx 66.21$ , with posterior median  $\text{Median}(m) = 65$ . The MAP estimate was approximately  $m_{\text{MAP}} \approx 65$ .

These results are also visually supported by the observed counts: the first part of the series contains mostly values between 1 and 5, while the second part exhibits a clear increase in the frequency of larger counts.

### 5.3 Separation Between the Two Rates

The posterior summaries indicate a substantial difference between the two Poisson intensities:

$$\lambda_1 \approx 3.21, \quad \lambda_2 \approx 5.11.$$

Moreover, the corresponding 95% credible intervals show very limited overlap.

This separation implies that the increase in the event rate is statistically significant according to the posterior distribution.

### 5.4 Consistency Between the Chains

Three independent Gibbs chains were initialized from substantially different starting values, but the traceplots showed that all chains converged toward the same posterior region and the initial transient phase disappeared after burn-in.

## 5.5 Autocorrelation and Effective Sample Size

Autocorrelation functions were computed for all parameters in order to assess the efficiency of the Markov chains.

The estimated effective sample sizes were extremely large, indicating that the autocorrelation structure of the chains is relatively weak.

This favorable behavior is likely due to the structure of the dataset itself, since the increase in the event rate is relatively pronounced and therefore the posterior distribution is well identified.

## 5.6 Why Hamiltonian Monte Carlo Cannot Be Applied Directly

Hamiltonian Monte Carlo (HMC) relies on gradient-based simulation of continuous trajectories in the parameter space. However, the change-point parameter  $m$  is discrete and so, it cannot be defined on the full parameter space  $(\lambda_1, \lambda_2, m)$ .

## 5.7 Final Conclusions

Overall, the Bayesian analysis strongly supports the presence of a change point in the observed count process, with the two Poisson regimes clearly separated, nearly at half of the dataset.